

Judging Nonlinear Equalizers at the FEC Threshold: A Zero-Multiplier Memoryless Inverse Clears KP4 Where a Neural Equalizer Does Not

Compliance-referred evaluation of PAM4 receiver equalization under front-end compression, with standard-cell implementation cost

Abstract—Neural-network (NN) equalizers are routinely motivated by front-end nonlinearity in high-loss wireline receivers. We show that this motivation, as usually evaluated, does not survive contact with three things a real link is defined by: its modulation (PAM4, not NRZ), its operating point (a forward-error-correction threshold, not an arbitrary bit-error-rate at which ratios are quoted), and its substrate (an ASIC, not a multiplier-free FPGA fabric). We rebuild the comparison on all three. First, we observe that a compressive front end acts on NRZ and PAM4 in categorically different ways: because $\tanh$ is odd and monotone, sign detection is invariant and the compression costs an NRZ slicer exactly 0 dB, whereas in PAM4 it closes the outer eye by up to 10.1 dB at the compression indices we study. The question "does a network earn its multipliers" is therefore only meaningful in PAM4. Second, we replace the ratio metric with a compliance-referred one: the SNR required to reach the KP4 RS(544,514) pre-FEC threshold, which we derive from the code parameters rather than quote (2.262×10^{-4} for a 10^{-15} post-FEC target, within 6 % of the value cited in practice), and we probe for an error floor *before* bisecting, so that a structure which can never comply is reported as such instead of returning a number. Third, we price implementations in standard-cell area read from the library's own LEF and SPICE views. At a compression index $\kappa=0.5$ on a 20 dB channel with a 7 bit ADC, an 11-tap FFE with an 8-tap DFE floors at 4.55×10^{-3} and is *non-compliant at any SNR*; so is a 96-multiplier MLP, which floors at 9.24×10^{-4} — $4.1 \times$ the threshold—on 4/4 seeds. A 128-entry, $896 \mu\text{m}^2$ lookup table implementing the exact memoryless inverse, costing *zero multipliers*, clears the threshold at 36.15 dB on 4/4 seeds. Adding that table to the network gives 36.25 dB, statistically indistinguishable from the table alone: the 96 multipliers are worth less than 0.1 dB once the free inverse is present, at $9.5 \times$ the silicon. Controls hold in both directions—on a linear channel the inverse does exactly nothing (36.10 = 36.10 dB) and the network is 0.25 dB *worse* than the linear baseline. We bound the claim: beyond $\kappa \approx 1$ every structure floors above the threshold, and the result is a statement about a modelled channel, not about a measured backplane, which we could not obtain. We also report ten measurement errors this work made and corrected, because each of them would have produced a publishable-looking number.

Index Terms—PAM4, wireline receivers, decision-feedback equalization, neural network equalizers, lookup tables, Reed–Solomon FEC, channel operating margin, standard-cell area, SkyWater sky130.

I. INTRODUCTION

A wireline receiver at 100 Gb/s per lane sees a channel that is strongly low-pass and an analog front end (AFE) that is not linear. The channel part is well understood and is handled by continuous-time linear equalization (CTLE), feed-forward equalization (FFE) and decision-feedback equalization (DFE). The front-end part—compression as the differential pair leaves its linear range—breaks superposition, and a linear equalizer cannot undo it by construction. This observation is the standard motivation for neural-network equalizers: the channel is nonlinear, so use a nonlinear learner.

The motivation is incomplete in a specific way. If the front-end nonlinearity is *memoryless*—if its output depends only on the present input—then its inverse is also memoryless, and after an B -bit ADC the signal takes only 2^B values. The inverse is therefore realizable *exactly* as a 2^B -entry table mapping one ADC code to another: no multipliers, no adders, no feedback. Whatever a network learns about a memoryless nonlinearity, that table already does exactly and nearly for free. A comparison that does not give the linear baseline this table is not measuring what the network contributes; it is measuring what compression costs.

This paper makes that comparison, and makes it at the place where the answer matters. Three methodological choices distinguish it from how such comparisons are usually reported.

Modulation. Compression is not a modulation-neutral impairment. Because $\tanh$ is odd and strictly increasing, $\text{sign}(\tanh(ax)) = \text{sign}(x)$ for all $x \neq 0$. NRZ detection is a sign decision, so—absent the equalizer—compression changes no NRZ decision at all; after automatic gain control the eye is unchanged, exactly 0 dB. PAM4 detection uses three thresholds, and compression pulls the outer levels inward while the inner eye actually widens, so the *minimum* distance, which sets the error rate, shrinks by up to 10.1 dB over the range we sweep (Section VII-D). Asking whether the memoryless inverse is worth anything in NRZ is asking at a point where the impairment is, for the detector itself, benign.

Operating point. A BER ratio is not a figure of merit unless the BER at which it is taken is stated, because the BER curve is exponential: in our own earlier measurements the same $7.98 \times$ ratio was worth 2.27 dB at 10^{-2} and 0.34 dB at 10^{-12} . We therefore report the SNR required to reach a threshold the standard fixes rather than a ratio at a point we choose. Modern 100/200 Gb/s-per-lane electrical links are not specified at a raw BER of 10^{-12} ; that requirement belongs to the pre-FEC era. They are specified around a pre-FEC BER near 2×10^{-4} with RS(544,514) "KP4" FEC delivering a post-FEC target near 10^{-15} . We derive that threshold from the code parameters (Section V-A) instead of quoting it.

Substrate. "Multiplier-free is cheaper" is true only where multipliers are built from fabric. On a device with hard DSP slices the ordering reverses, as we measured previously and

repeat here. But a 112G/224G SerDes is an ASIC, so the question that matters is silicon area, which we obtain by synthesizing to generic gates and pricing each gate with the area recorded in the SkyWater sky130 standard-cell library's own LEF view, with transistor counts from its SPICE netlists.

A. Contributions

- **A compliance-referred evaluation protocol** for receiver equalizers: derive the FEC threshold from the code, probe for an error floor before searching for a required SNR, and refuse to report a required SNR for a structure that floors above the threshold (Section V).
- **The first measurement, at a standard's own threshold, of what a neural equalizer contributes over the free memoryless inverse.** The answer is under 0.1 dB, with the sign not even consistent, at $9.5\times$ the silicon (Section VII-B, VIII-A).
- **A decisive negative result:** a 96-multiplier MLP is non-compliant at the operating point where a zero-multiplier 128-entry table is compliant, on 4/4 seeds (Section VII-B).
- **An exact statement of why the modulation matters:** compression costs NRZ detection exactly 0 dB and PAM4 up to 10.1 dB, with the arithmetic given (Section VII-D).
- **Standard-cell cost** for all arms read from the PDK rather than estimated, together with the FPGA-family reversal, so that the substrate dependence is visible rather than hidden (Section VIII).
- **Ten corrected measurement errors**, reported rather than silently fixed, each of which produced a plausible number before it was caught (Section X).

II. RELATED WORK, AND WHY THIS EVALUATION IS NEEDED

A. Linear equalization in wireline links

The standard receiver chain—CTLE, then a sampled FFE, then a DFE—is textbook [1] and its wireline specialization is treated in [17]. The division of labour is structural rather than conventional: a DFE can only cancel post-cursor intersymbol interference, because it feeds back decisions that have already been made, so pre-cursor ISI is necessarily the FFE's. The DFE's advantages (it does not amplify noise) and its costs (error propagation; a feedback loop that must close within one unit interval) are equally structural. We adopt this chain unchanged as the baseline and as the substrate on which all nonlinear arms are added.

B. Sequence detection and achievable bounds

Symbol-by-symbol equalization is not optimal on an ISI channel; maximum-likelihood sequence estimation is [6], at a state count exponential in the memory. Reduced-state variants—RSSE [8] and delayed decision-feedback sequence estimation [7]—trade states for per-survivor decision feedback. We use a calibrated matched-filter bound together with a DDFSE implementation not as a proposal but as a *ceiling*: without it, a comparison among suboptimal structures cannot say whether any of them is close to what the channel permits. Our earlier measurements placed every structure

considered here, including our own, at least $25\times$ from that ceiling, except in one narrow regime (Section IX-C).

C. Nonlinear equalization before neural networks

Nonlinear equalization of channels with memory has a long history in Volterra-series form [10], [11], where the parameter count grows combinatorially with kernel order and memory. Look-up architectures are equally old: distributed arithmetic [13], [14] replaces the multipliers of an inner product with a ROM addressed by a bit-slice across the data words. It is worth being precise about the relationship, because the two are often conflated. Distributed arithmetic's ROM stores a *linear* function of its address—it is an exact rearrangement of a multiply–accumulate, and a fitted linear model reproduces its contents to numerical precision. The table proposed here stores a function that is not linear in its address; that is the whole point of it.

D. Neural equalizers, and how they are evaluated

A large recent literature applies small multilayer perceptrons, and latterly recurrent and convolutional structures, to wireline and short-reach optical receivers, generally reporting a BER improvement over an FFE/DFE baseline and a parameter or multiplier count [refs to be supplied—see Section R]. Three features of that evaluation practice recur and are what this paper responds to:

- The baseline is a *linear* equalizer with no nonlinear front-end correction of any kind. If the dominant impairment is memoryless, this attributes to the network a gain that a table obtains for free.
- The comparison is reported as a BER ratio at a BER between 10^{-2} and 10^{-4} , which is not the operating point of a FEC-protected link and, being a ratio on an exponential curve, does not transfer to one.
- Cost is reported as parameter count, multiplier count, or FPGA fabric utilization, none of which is silicon area on the substrate the application actually uses.

E. The standardized operating point

Ethernet electrical interfaces at 50 Gb/s per lane and beyond are specified with RS-FEC: RS(528,514) "KR4" and RS(544,514) "KP4" over GF(2^{10}) [4], [5], with KP4 correcting $t = 15$ symbols per codeword [9]. Channel compliance itself is expressed through Channel Operating Margin [3], whose internal detector-error-ratio target is of the same order as the pre-FEC BER used here. That the standard fixes a threshold is what makes a threshold-referred comparison possible at all; it removes the freedom to choose the operating point that makes a ratio look large.

To our knowledge no published comparison of a neural equalizer against a linear baseline in a compressive receiver (i) gives the baseline the zero-multiplier memoryless inverse that the impairment's own structure makes available, (ii) judges both at a FEC threshold derived from the code rather than at a chosen BER, and (iii) prices both in standard-cell area. Each omission individually flatters the network. Taken together they can invert the conclusion, which is what we find.

III. SYSTEM MODEL

A. Modulation

Symbols are drawn uniformly from an M -PAM alphabet normalized so that the outer levels are ± 1 ,

$$b \in \{ (2k - (M-1)) / (M-1) \}, \quad k = 0 \dots M-1 \quad (1)$$

so that $M=2$ gives ± 1 and $M=4$ gives $\{-1, -1/3, +1/3, +1\}$. This normalization keeps peak amplitude constant across modulations, so the two are compared at equal peak power and the textbook penalty appears without adjustment: the minimum distance falls from 2 to $2/3$, giving

$$20 \log_{10}(3) = 9.5424 \text{ dB} \quad (2)$$

Bits are Gray-mapped, so adjacent-level errors—which are essentially all errors at the error rates of interest—produce one bit error each. Two invariants of this mapping are used as self-checks throughout: the measured symbol error rate matches $2(1-1/M) Q(d/2\sigma)$ to $1.026\times$ over 752 events, and BER/SER = 0.5 exactly.

One consequence of (1) is easy to overlook and cost us a spurious error floor (Section X-A): $E[b^2]$ is 1 for NRZ but $5/9$ for PAM4. Any gain estimate formed as a correlation $E[xb]$ is the cursor scaled by $E[b^2]$, so it must be normalized by it. In NRZ the normalization is the identity, which is exactly why the error is invisible until the modulation changes.

B. Channel

The channel magnitude uses the insertion-loss fitting form that IEEE 802.3 uses to describe measured channels, so that the model is stated in the same coordinates as a specification:

$$\text{IL}(f) [\text{dB}] = a_0 + a_1 \sqrt{f} + a_2 f + a_4 f^2 \quad (3)$$

whose terms correspond to frequency-independent connector and bonding loss, conductor skin effect, dielectric loss, and surface roughness. A via-stub resonance is applied multiplicatively as an open stub of electrical delay τ , which produces a notch no power of f can reproduce and which is the feature that most often separates equalizers on measured channels. Phase is not free: a linear-phase reconstruction gives a symmetric cursor set and removes the post-cursor asymmetry a DFE exists to cancel, so the minimum-phase response is built from the real cepstrum. Two further channel constructions are supported and used for cross-checks—one built from transmission-line geometry and materials through an independent library [23] with a causal dielectric model [19]

and microstrip dispersion [20], and one built by cascading a measured two-port. Neither is a measured backplane; see Section XI.

C. Impairments

Noise is added after the channel and before the receiver-side impairments, so that the data-processing inequality applies to everything downstream: no receiver impairment can increase the information available to the detector. Compression is applied at the AFE position,

$$y = \tanh(a u)/a, \quad a = \kappa/\text{rms}(u) \quad (4)$$

with the compression index κ equal to the ratio of input rms to the pair's linear range. The model also provides slew-rate limiting, AC-coupling droop, far-end crosstalk, multiplicative supply-gain modulation, sampling jitter, and a post-compression band limit; these were used to search for a regime in which nonlinearity acquires memory (Sections IX-A, X). A B -bit ADC follows, with full scale set to 2.5σ of the AGC-normalized signal.

IV. THE MEMORYLESS INVERSE IS A TABLE

PROPOSITION 1. *Let f be memoryless and strictly monotone on the signal range, and let the received sample be quantized to B bits. Then f^{-1} is realizable exactly by a 2^B -entry table mapping one ADC code to another, requiring no multipliers, no adders, and no feedback.*

Proof. Memorylessness gives $y[n] = f(u[n])$, so $u[n] = f^{-1}(y[n])$ depends on the present sample only; a function of one variable is memoryless. Quantization restricts the domain of that function to a finite set of cardinality 2^B . A function on a finite domain is a table. $\square$

Three remarks make the proposition operational rather than decorative.

It is exact, not approximate. Within the quantizer's own resolution the table commits no modelling error, because it is not a model: it is the function. A learner approximating the same function incurs approximation error that no amount of training removes, only reduces. This distinction is what Section VII-B measures, and it is the reason the network floors where the table does not.

Its cost does not scale with the nonlinearity's severity. A 7 bit ADC gives 128 entries whether κ is 0.1 or 3. Approximation cost, by contrast, grows with curvature.

It is an upper bound on all memoryless correction. Any front end that corrects the compression sample-by-sample— analog predistortion, a polynomial, a small network restricted to one tap—is a memoryless function of the same quantized variable, hence a table, hence no better than this one. Giving the baseline this table is therefore not a courtesy; it is the only way to isolate what a structure contributes *beyond* memorylessness.

The converse delimits the claim. If the nonlinearity has memory—if the compression is followed by filtering that mixes compressed samples, or if it has internal state as slew limiting does—then the inverse has memory too, and Proposition 1 does not apply. Whether such a regime exists at

physically reachable parameter values is an empirical question, addressed in Section IX-A.

V. COMPLIANCE-REFERRED METHODOLOGY

A. Deriving the threshold rather than quoting it

KP4 is RS(544,514) over GF(2^{10}), correcting $t = 15$ symbols per 5440-bit codeword. For a bounded-distance decoder, a codeword is corrected if and only if at most t of its n symbols are in error, so the residual symbol error rate after decoding is

$$P_{\text{out}} = \sum_{j>t} (j/n) C(n,j) p_s^j (1-p_s)^{n-j} \quad (5)$$

with $p_s = 1-(1-p_b)^{10}$ the symbol error rate implied by a bit error rate p_b . Evaluating (5) term by term at $n=544$ loses precision; the identity $j C(n,j) = n C(n-1,j-1)$ folds it to a single tail probability,

$$P_{\text{out}} = p_s \Pr[\text{Bin}(n-1, p_s) \geq t] \quad (6)$$

and the post-FEC bit error rate follows by scaling with the observed number of bad bits per bad symbol. Bisection (6) for a post-FEC target of 10^{-15} gives a pre-FEC threshold of 2.262×10^{-4} , within 6 % of the figure used in practice. The same computation with KR4's $t=7$ returns 2.182×10^{-5} , confirming that the number responds to the code rather than being a constant we inserted. A unit test compares the folded form against a brute-force sum on a small code.

B. Floor-first: refusing to report a number that does not exist

Bisection for "the SNR at which a structure reaches the threshold" returns a number even when the structure has an error floor above the threshold and reaches it at no SNR whatsoever. That number is then read as a result. We therefore run a single high-SNR probe (55 dB) before bisection. If the BER there still exceeds the threshold the arm is reported as floored and *no required SNR is produced*. Zero observed errors is not a BER of zero: it is treated as an upper bound $3/N$, and if that bound itself exceeds the threshold the run is reported as having too few symbols rather than as a pass.

C. Checking the assumption the threshold rests on

The threshold in Section V-A assumes errors are independent across RS symbols. DFE error propagation violates this by clustering errors—and does so in the *favourable* direction, since the same bit errors packed into fewer symbols are easier for a symbol-correcting code. We therefore measure the per-codeword symbol-error histogram and compare its variance against the binomial prediction; a ratio outside $[0.5, 1.5]$ flags the post-FEC figure as not trustworthy rather than silently reporting it. As an illustration of the size of the effect, 4000 bit errors scattered at random corrupt 3692 RS symbols, while the same 4000 packed in runs of ten corrupt 400.

D. Statistical discipline

All comparisons are run across independent seeds and reported per seed, with the count of seeds on which each arm reached the threshold, because a mean over seeds can be carried by one. Training and measurement windows are disjoint (first 30

% trains taps, tables and network weights; BER is counted only on the remainder). Symbol alignment is measured from the pulse response rather than assumed.

VI. EXPERIMENTAL SETUP

Unless stated otherwise: PAM4, 20 dB Nyquist insertion loss, 8 samples per symbol, 7 bit ADC at 2.5σ full scale, 200 000 symbols (400 000 bits) per run, 30 % training fraction, four seeds. The four arms are:

- **Linear baseline**—11-tap NLMS FFE and 8-tap DFE. The NLMS step is normalized by both signal power and tap count; normalizing by power alone diverges at 29 taps (Section X-J).
- **+ memoryless inverse**—the same baseline with a 128-entry code-to-code table ahead of the FFE. Zero multipliers.
- **MLP, 96 multipliers**—an 11-input, 8-hidden-unit, single-output perceptron with tanh activation, replacing the FFE, followed by the same 8-tap DFE. Trained with Adam on MSE for 8 epochs; convergence is checked and non-convergence is reported rather than scored as a BER.
- **MLP + memoryless inverse**—both.

A fifth arm, a 256-entry sample-indexed table which was the proposal of our earlier work, is carried through the sweeps for continuity and is reported where it is informative.

VII. RESULTS

A. Controls

On a linear channel ($\kappa=0$) the optimum equalizer is linear, so nothing should win. The memoryless inverse does exactly nothing, to the digit: 36.10 dB against 36.10 dB. The sample-indexed table costs 7.73 dB, as it must, since two quantized samples cannot span what eleven FFE taps and eight DFE taps span. The MLP is 0.25 dB worse than the linear baseline, which is the expected cost of fitting a nonlinear approximator where the optimum is linear. Any arm winning here would indicate a leak in the harness.

B. The compliance point

Table I is the paper's result.

Table I.

COMPLIANCE AT $\kappa=0.5$ (PAM4, 20 dB, 7 BIT ADC, KP4 THRESHOLD 2.262×10^{-4} , FOUR SEEDS). "FLOOR" IS THE BER STILL PRESENT AT 55 dB SNR; AN ARM THAT FLOORS ABOVE THE THRESHOLD REACHES IT AT NO SNR. AREA IS STANDARD-CELL SUM IN SKY130 (SECTION VIII-A).

Arm	Multipliers	Compliant	Floor / required SNR	Seeds	Area (μm^2)
Linear FFE11 + DFE8	11	no	floor 4.55×10^{-3} (20.1× thr.)	0/4	37 666
+ memoryless inverse	0 added	yes	36.15 dB	4/4	38 562
MLP (96 mult.)	96	no	floor 9.24×10^{-4} (4.1× thr.)	0/4	368 026
MLP + memoryless inverse	96	yes	36.25 dB	4/4	368 922

Per-seed values do not straddle the boundary. The inverse reaches the threshold at 35.88, 36.25, 36.79 and 35.68 dB; the network floors at 8.2, 4.4, 14 and 10×10^{-4} . The outcome is unanimous in both directions.

Three findings follow, and it is worth separating them because they are often conflated.

The network does learn the inverse—partially. Its floor is $4.9 \times$ below the linear baseline's. A structure that had learned nothing about the compression would floor where the baseline does. The network is doing the right thing; it is doing it approximately, and the approximation error is large compared to a threshold that sits four times below its floor.

Once the exact inverse is present, the multipliers buy nothing measurable. 36.25 dB with the network against 36.15 dB without it—a difference of 0.10 dB, against a seed-to-seed spread of 0.62 and 1.11 dB in the two arms. The sign is not even consistent with the network helping; the table alone is nominally *lower*. The honest statement is that 96 multipliers are worth less than 0.1 dB here, not that they are worth -0.1 dB.

The cost ratio is $9.5 \times$. Section VIII-A.

C. Range of validity

Beyond $\kappa \approx 1$ every arm floors above the threshold: at $\kappa=1.0$ the floors are 6.2×10^{-2} (linear), 5.7×10^{-4} (inverse) and 1.2×10^{-3} (sample table); at $\kappa=2.0$ they are 1.8×10^{-1} , 4.8×10^{-2} and 4.4×10^{-2} . Compression that severe destroys information that no receiver structure recovers, and the claim of this paper ends there. We state the boundary rather than sweeping only inside it.

D. Why the modulation decides whether the question is meaningful

For NRZ the statement is exact. *tanh* is odd and strictly increasing, so

$$\text{sign}(\tanh(ax)) = \text{sign}(x), \quad x \neq 0 \quad (7)$$

and NRZ detection is precisely a sign decision. After AGC restores the levels to ± 1 the eye is unchanged: compression costs the NRZ slicer **0.00 dB**, not approximately but identically. Compression can still hurt an NRZ *link*, but only through the equalizer, whose linearity assumption it breaks—not through the detector.

For PAM4 the three thresholds sit at different points on the compressive curve, whose slope decreases with $|u|$. The outer level spacing is compressed more than the inner. Table II gives the arithmetic after AGC.

Table II.

EYE OPENINGS AFTER COMPRESSION AND AGC, PAM4. THE INNER EYE *WIDENS*; THE OUTER EYE *CLOSES*; THE MINIMUM DISTANCE, WHICH SETS THE ERROR RATE, FOLLOWS THE OUTER EYE. THE CORRESPONDING NRZ COLUMN WOULD READ 0.00 dB AT EVERY ROW.

κ	levels after AGC	inner eye	outer eye	Δ min. dist.
0	$\pm 1, \pm 0.333$	0.667	0.667	—
0.5	$\pm 0.993, \pm 0.355$	0.710	0.638	-0.38 dB
1.0	$\pm 0.974, \pm 0.411$	0.822	0.563	-1.47 dB
1.5	$\pm 0.950, \pm 0.485$	0.970	0.465	-3.13 dB
2.0	$\pm 0.925, \pm 0.559$	1.118	0.366	-5.22 dB
3.0	$\pm 0.885, \pm 0.678$	1.355	0.208	-10.13 dB

This is the justification for rebuilding the evaluation in PAM4 rather than a matter of following current practice. In NRZ the impairment under study is, for the detector, invisible.

VIII. IMPLEMENTATION COST

A. ASIC standard cells

Each design is synthesized to a generic gate set and every gate priced with the area recorded in the sky130 high-density library's own LEF SIZE record, with transistor counts from the library's SPICE netlists [24]. No area constant is written by us. Flip-flops are mapped to the library's asynchronous-reset cell; the enable-plus-reset variant, which has no single cell, is priced as its actual implementation (reset flop plus a 2:1 multiplexer). All drive strengths are fixed to the smallest, which is an assumption, but one shared by every arm, so the ratio is insensitive to it.

Table III.

STANDARD-CELL COST, SKY130. AREAS ARE CELL-AREA SUMS, NOT POST-PLACE-AND-ROUTE DIE AREA, AND MUST BE READ AS RATIOS. THE COMPLIANT STRUCTURE IS THE LINEAR BASELINE PLUS THE 128-ENTRY INVERSE TABLE.

Design	gates	area (μm^2)	transistors	ratio
FFE11 + DFE8 (linear)	4 778	37 666	41 052	1.000
+ inverse ROM (128×7)	112	+896	944	1.024
MLP 11×8 (96 multipliers)	51 585	368 026	401 706	9.77
MLP 5×2 (small)	6 317	46 545	50 870	1.236
FFE11 + sample table (256)	4 802	42 735	46 408	1.135

The compliant structure costs **2.4 % more silicon** than the non-compliant baseline. The 96-multiplier network costs **9.5× the compliant structure** and does not comply. Note that the ROM's cost cannot be measured with a placeholder table: synthesizing an identity map yields $175 \mu\text{m}^2$ and 7 gates

because the synthesizer resolves it to wires. The figure above uses the actual de-compression coefficients (Section X-B).

B. FPGA, and why the substrate inverts the answer

On iCE40, which has no hard multipliers, the table-based equalizer costs 3 037 logic cells at 41.4 MHz against 2 999 cells at 34.7 MHz for the linear baseline—+1.3 % area at 1.19× the clock. On Xilinx 7-series the ordering reverses: the table costs 1 316 fabric cells while the network costs 391 cells plus 12 DSP slices, and the DSP slices are already on the die. We report both, and we regard the reversal as the substantive point rather than an inconvenience: *multiplier-free is an algorithmic property, not a cost property*, and it converts into cost only where multipliers consume the same resource as logic. What makes the ASIC number decisive for this application is simply that a 112G/224G SerDes is an ASIC.

IX. VALIDATION AGAINST THEORY AND DEVICES

A. The nonlinearity-with-memory regime, and a refuted hypothesis

Proposition 1 fails when compression acquires memory, so we searched for that regime. The hypothesis was that a long-memory impairment would break the table and make a network necessary. It was not supported. Ten octaves of AC-coupling droop (τ up to 18.8 UI) moved the network-to-table ratio only from 1.28 to 1.47, whereas the compression severity moved it from 1.28 to 3.40. Slew-rate limiting produced no usable window at all: mild limiting acts as a low-pass and improves BER, while severe limiting drives the output toward the integral of the sign sequence and destroys information for every structure alike. The binding variable is the severity of the nonlinearity, not its memory. We report this because the hypothesis was ours and it was wrong.

B. The decision-domain ceiling

A structure indexed by past *decisions* cannot exceed a ceiling set by what decisions discard. Measured in the band-limited regime: a linear DFE 1.00×, a truncated Volterra expansion over ± 1 decisions 1.04–1.10× at order 2 (79–137 parameters) and 2.68–2.91× at order 3 (299–1351 parameters), and a fully populated decision-indexed ROM 3.08–3.09× (256–1024 entries). Growing the parameter count by 17× moves nothing, because the limitation is the index, not the capacity: past decisions have already thrown away amplitude. This is why the useful table is indexed by ADC codes rather than by decisions.

C. Detection bounds

A calibrated matched-filter bound and an achievable reduced-state sequence detector place every structure considered here—ours included—at least 25× from what the same channel and noise permit, with one exception: under post-compression band limiting at a 3 dB bandwidth near $0.35/T$, the table reaches 2.3×10^{-3} against 2.2×10^{-2} for the best linear-model sequence detector. Reporting the bound is what prevents a comparison among suboptimal structures from being mistaken for a statement about optimality.

D. Circuit-level validation of the model's knobs

The simulator's knobs are only meaningful if they correspond to circuit quantities. Each was checked against SPICE on the sky130 process [24], [25], and three of the closed forms we had written were found to be stated without their range of validity.

Compression index. V_{lim} is not the weak-inversion constant $2nV_T \approx 67$ mV but runs 78–297 mV with bias, so a translation of κ into millivolts must carry its bias point.

Slew. A correctly sized current-mode-logic driver at 20 GBd has a 20–80 % edge of 0.064–0.071 UI, and its single-ended swing equals IR_{eff} to a ratio of 1.000 when the tail current is measured through a separate sense source—a consistency check that closes Kirchhoff's law along a second path. The closed form $\tau = R_{eff} C_L$ is accurate to 0.2–1.2 % once C_L dominates, with a 2.33 ps floor set by the differential pair's own switching that the closed form omits. Inverting the relation, the slew values swept in our earlier work correspond to load capacitances of 5.0–16.7 pF, against 0.1–0.3 pF for a real pad, ESD and package—a factor of 20–170. That sweep was therefore never a driver; it was a measure of how far one must go to break the table, and consistently with that, slew never broke it.

Comparator regeneration. A StrongARM latch's regeneration time constant, extracted three independent ways (a 2000× input sweep against the logarithmic resolve law, the slope of $\ln|v_{od}|$ on the waveform, and a small-signal pole measurement), agrees to 3 %: 16.6–17.2 ps. The closed form $\tau = C_L/g_m$ is correct, but C_L is the capacitance the regenerative loop carries—80.2 fF including the cross-coupled gates—not the 2 fF external load; $80.2 \text{ fF} / 4.699 \text{ mS} = 17.07 \text{ ps}$ against 16.85 ps measured. The consequence is a scope condition on a claim we had made unconditionally: at 20 GBd one UI is only 2.97τ , giving a sensitivity of 96 mV, so "quantization rather than the comparator sets the floor" holds only at three-way interleaving or beyond. Real ADC-based receivers interleave far more than that, but the condition belongs in the statement.

Offset. Monte-Carlo mismatch gives an input-referred offset of $\sigma = 1.4$ –2.0 mV for the input pair alone but **4.06 mV for the whole latch** (20 seeds, bisection to 25 μ V), which is 1.55× the 7 bit ADC's 2617 μ V LSB and therefore the largest of the three floor contributors. Pelgrom's law [15] is verified, with area equal to WL .

X. ERRORS FOUND AND CORRECTED

Each of the following produced a plausible number before it was caught. We list them because a methods section that reports only the procedures that worked describes a study that was never at risk.

- **A. AGC without $E[b^2]$.** Estimating cursor gain as the correlation $E[xb]$ overstates it by 9/5 in PAM4. Symptom: a BER floor at 5×10^{-2} that persisted to 55 dB SNR with no impairment present—a *scale* error that would have been read as an impairment floor.
- **B. ROM area measured with an identity table.** 175 μm^2 and 7 gates, because the synthesizer removed it. The real

coefficients give $896 \mu\text{m}^2$ and 112 gates.

- **C. Unmapped gates omitted from an area sum.** Enable-plus-reset flip-flop variants had no mapping entry and were silently skipped, biasing area downward—in the direction that flatters the table. The tool now refuses to report an area when any gate is unmapped.
- **D. Comparator resolve read on a single node.** Measuring the latch on one output caught the common-mode fall, not the differential resolve, producing a false sensitivity plateau at 47.3 ps. Tightening solver tolerances by $1000\times$ did not move it. A zero-input run exposed it: the latch did not resolve at all (both outputs at 0.867 V, metastable) while the measurement still returned a time.
- **E. Swing read as a window extremum.** Taking MAX/MIN of the differential output over a window captured the input edge's C_{gd} feedthrough spike and exceeded the ideal $2IR_{\text{eff}}$ bound by 30 %—the same symptom that had previously been diagnosed as a broken design. Reading at a settled instant gives a ratio of 1.000.
- **F. Device width semantics.** In sky130 w is total width and n_f partitions it; our devices were $20\text{--}40\times$ undersized, which drove a tail node to -0.337 V—an ideal current source demanding a current the device could not deliver. Three earlier driver attempts were abandoned as "unmeasurable" for this reason before the cause was found.
- **G. Pelgrom area as $W L n_f$ rather than $W L$,** inflating reported areas $4\times$. Ratios and conclusions were unaffected, which is precisely why it survived.
- **H. A noise-gain expectation evaluated at the wrong point.** $E[\cosh^4]$ for the de-compression noise gain predicted 16 dB against $5.3\times$ measured, because the mean is dominated by rare large excursions; evaluating at the decision amplitude gives 7.5 dB.
- **I. Sequence-detector bookkeeping.** A sign error in trellis alignment and a double-cancellation in per-survivor feedback each pinned the bound at a BER that no SNR improved—a "bound" that was in fact a bug.
- **J. NLMS step normalized by power alone.** Stable only while $\mu < 2/(L\sigma^2)$; at 29 taps the coefficients diverged to 10^6 . Concluding "more taps do not help" from that state measures the step size, not the channel.

Three of these (A, C, E) bias results in the direction of the paper's own thesis. We regard that as the reason to report them.

XI. LIMITATIONS AND THREATS TO VALIDITY

- **No measured backplane.** This is the principal limitation. The channels are an 802.3-form insertion-loss fit, a geometry-and-materials transmission-line construction, and a cascade of a measured two-port—none of which is a backplane measurement. We were unable to obtain Touchstone data for one. The claim is therefore about a modelled channel class, not about a product.
- **One network, one training budget.** The MLP is 11×8 , 8 epochs, 200 000 symbols. A larger or longer-trained network may clear the threshold. The claim is *not* that

networks cannot; it is that this one costs $9.5\times$ the silicon and does not, while the exact inverse costs $896 \mu\text{m}^2$ and does—and that a larger network moves the area ratio further in the same direction.

- **Cell-area sums, not die area.** Table III omits place-and-route utilization; only ratios are claimed.
- **Single process node.** sky130 is a 130 nm process. The regeneration and offset figures in Section IX-D are statements about it, not about the 5–7 nm nodes of production SerDes, where τ is an order smaller.
- **Compression model.** A single *tanh* stands for the AFE's compression. Real front ends have several compressive stages with memory between them; where that memory is significant, Proposition 1 weakens and the network's case strengthens.
- **Post-FEC figures are conditional.** They assume independent symbol errors; the harness measures the clustering and flags the assumption when it fails, but flagged figures remain estimates.

XII. FUTURE WORK

Immediate and decisive. Repeat Table I on a measured backplane. The entry point exists: any Touchstone file drops into the same pipeline. Until then the result's channel scope is a modelling assumption.

Close the remaining standards gap. Implement Channel Operating Margin [3] in full, so that channels are admitted by the same criterion a compliance body would use, and report margin in the units a working group reads.

Parallelism as a cost axis. A 20 GBd receiver on 500 MHz logic needs ~ 40 -way unrolling, so 12 DSP slices become ~ 480 and exceed mid-range 7-series capacity. Our FPGA comparison is per-lane and single-rate; redoing it at deployment parallelism would test whether the substrate reversal of Section VIII-B survives at scale. We flag this as a hypothesis we have *not* measured.

Find the memory regime honestly. Proposition 1's converse is the paper's real boundary. A front-end model with compression interleaved between filter stages—rather than followed by one—is the natural place for a network to earn its multipliers, and our earlier band-limiting experiments hint at it. That regime should be sought in a physically parameterized AFE, not by sweeping an abstract knob until something breaks.

Offset calibration in the loop. The largest floor contributor is comparator offset at 4.06 mV, above the ADC LSB. Production receivers calibrate it; the model does not. Adding calibration would lower every arm's floor and might move the κ boundary of Section VII-C outward, which would enlarge the region in which the paper's claim holds.

Stronger codes. 200 Gb/s-per-lane work is considering concatenated FEC with different pre-FEC targets. The methodology transfers unchanged—the threshold is derived from the code—so the comparison can simply be re-run against a different n, k, t .

XIII. CONCLUSION

A compressive front end in an ADC-based PAM4 receiver presents a nonlinearity whose inverse, being memoryless, is exactly a 2^B -entry table costing no multipliers. Evaluated at the KP4 pre-FEC threshold derived from the code, that table moves a link from non-compliant at any SNR to compliant at 36.15 dB, on four seeds out of four, for 2.4% additional standard-cell area. A 96-multiplier neural equalizer at the same operating point is non-compliant, and once the table is present its multipliers are worth less than 0.1 dB at $9.5\times$ the silicon. The result does not show that neural equalizers are useless; it shows that a comparison which omits the free memoryless inverse, reports ratios away from the standardized operating point, and prices multipliers on a fabric that has none, can invert its own conclusion on all three counts at once.

APPENDIX: REPRODUCTION

REPRODUCING TABLE I

```
import comply, nneq
NN = {"dorigi": nneq.link, "_ppaelgeot": ("FFetap",),
      "window": 5, "hidden": 8, "epochs": 8}
base = dict(symbols=200000, lossdB=20.0, FFetap=11, DFetap=8,
           ADCbits=7, levels=4, compression=0.5)
arms = {"linear": {}, "+inverse": dict(decompress=True),
        "MLP96": dict(NN), "MLP96+inv": dict(NN, decompress=True)}
print(comply.말로(comply.팔들(base, arms, seeds=(0,1,2,3))))
```

NOTE. The repository's identifiers are Korean; the listing above transliterates them for readability. Threshold derivation is fec.문턱(); the floor-first protocol is comply.필요SNR(); standard-cell pricing is asic.잔것(); circuit measurements are in afespice.py. Channel construction, impairments and the PAM4 primitives are in serdes.py and pam.py.

REFERENCES

EDITORIAL NOTE — REMOVE BEFORE SUBMISSION

This bibliography was assembled without access to any bibliographic database: arXiv, Crossref, Semantic Scholar, DBLP, OpenAlex, DOI resolution and IEEE Xplore were all unreachable from the environment in which this draft was produced. **Not one entry has been verified against a record.** Each is therefore marked with its confidence, and fields that could not be recalled with confidence are left as *[verify]* rather than filled with a plausible guess. Section R lists what must be added. A fabricated citation is worse than a missing one, so nothing here is invented to look complete.

CONFIDENCE KEY. ♦ content and bibliographic fields both recalled with high confidence · ◇ content confident, fields to be confirmed · ○ placeholder: the citation is required at this point but must be selected and verified.

A. Equalization and detection

1. ♦ J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. New York: McGraw-Hill. — CITED FOR: the FFE/DFE/MLSE framework, the Q-function error expressions, and the M-PAM symbol-error formula used as a self-check in Section III-A.
2. ◇ E. A. Lee and D. G. Messerschmitt, *Digital Communication*, Kluwer. — Adaptive equalization and timing recovery; LMS/NLMS stability conditions referenced in Section X-J.
3. ◇ IEEE Std 802.3, Annex 93A, "Channel Operating Margin (COM)." *[verify clause and edition year]* — The standardized channel figure of merit; cited in Sections II-E and XII as the criterion this work approximates but does not yet implement.
4. ◇ IEEE Std 802.3bj, "Physical Layer Specifications for 100 Gb/s Operation Over Backplanes and Copper Cables." *[verify year]* — Introduces

the RS-FEC variants; source for the KR4 RS(528,514) and KP4 RS(544,514) parameters.

5. ◇ IEEE Std 802.3ck, "Physical Layer Specifications for 100 Gb/s, 200 Gb/s and 400 Gb/s Electrical Interfaces." *[verify year]* — The 100 Gb/s per-lane PAM4 electrical interface whose operating point Section V-A targets.
6. ♦ G. D. Forney, Jr., "Maximum-likelihood sequence estimation of digital sequences in the presence of intersymbol interference," *IEEE Trans. Inf. Theory*, vol. 18, no. 3, pp. 363–378, May 1972.
7. ◇ A. Duel-Hallen and C. Heegard, "Delayed decision-feedback sequence estimation," *IEEE Trans. Commun.*, 1989. *[verify volume, issue, pages]* — The reduced-state detector with per-survivor tail cancellation implemented as the achievability bound in Section IX-C.
8. ◇ M. V. Eyuboglu and S. U. H. Qureshi, "Reduced-state sequence estimation with set partitioning and decision feedback," *IEEE Trans. Commun.*, 1988. *[verify volume, issue, pages]*
9. ♦ I. S. Reed and G. Solomon, "Polynomial codes over certain finite fields," *J. Soc. Ind. Appl. Math.*, vol. 8, no. 2, pp. 300–304, 1960. — The code family whose bounded-distance decoding gives (5)–(6).

B. Nonlinear equalization

10. ○ Volterra-series equalization of nonlinear channels. *[select and verify — the canonical satellite/voiceband nonlinear-ISI references, e.g. work by Benedetto and Biglieri, and by Biglieri, Gersho, Gitlin and Lim in the early 1980s]* — Cited in Section II-C for the combinatorial parameter growth that motivates truncation, and in Section IX-B where a truncated decision-domain Volterra expansion is one of the structures measured against the ceiling.
11. ○ Adaptive cancellation of nonlinear intersymbol interference. *[select and verify]*
12. ○ **Neural-network equalizers for wireline and short-reach optical receivers.** *[This is the most important gap. Section II-D characterizes an evaluation practice — linear-only baseline, BER ratio away from the FEC point, parameter/multiplier cost — and that characterization must be supported by three to six representative and recent citations that the authors have read. Candidates to search: MLP/CNN/RNN equalizers in IEEE JLT, OFC/ECOC proceedings, IEEE TCAS-I/II, and DesignCon, 2019–2025. Any claim in Section II-D that a cited work does not in fact make must be withdrawn.]*

C. Multiplier-free arithmetic

13. ♦ A. Peled and B. Liu, "A new hardware realization of digital filters," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 22, no. 6, pp. 456–462, Dec. 1974. — Distributed arithmetic. Cited in Section II-C for the distinction that matters here: a DA ROM stores a function *linear* in its address, whereas the table of Proposition 1 does not.
14. ◇ S. A. White, "Applications of distributed arithmetic to digital signal processing: A tutorial review," *IEEE ASSP Magazine*, 1989. *[verify volume, issue, pages]*

D. Circuits and devices

15. ♦ M. J. M. Pelgrom, A. C. J. Duinmaijer and A. P. G. Welbers, "Matching properties of MOS transistors," *IEEE J. Solid-State Circuits*, vol. 24, no. 5, pp. 1433–1439, Oct. 1989. — The $1/\sqrt{WL}$ mismatch law verified in Section IX-D.
16. ♦ B. Razavi, "The StrongARM latch [A circuit for all seasons]," *IEEE Solid-State Circuits Magazine*, vol. 7, no. 2, 2015. *[verify pages]* — Regeneration rather than amplification; the $\tau = C_L/g_m$ form whose C_L Section IX-D corrects.
17. ◇ B. Razavi, *Design of Integrated Circuits for Optical Communications*. — CML drivers, CTLE, and the high-speed receiver chain of Section III.
18. ○ Origin of the clocked sense-amplifier latch topology. *[select and verify — commonly attributed to a 1990s JSSC sense-amplifier paper; confirm before attributing]*

E. Channel and material models

19. ◇ A. R. Djordjević *et al.*, "Wideband frequency-domain characterization of FR-4 and time-domain causality," *IEEE Trans. Electromagn. Compat.*, 2001. *[verify volume, issue, pages, full author list]* — The causal dielectric model used by the geometry-based channel construction.

20. ◇ M. Kirschning and R. H. Jansen, "Accurate model for effective dielectric constant of microstrip with validity up to millimetre-wave frequencies," *Electronics Letters*, 1982. [verify volume, pages]
21. ◇ E. Hammerstad and Ø. Jensen, "Accurate models for microstrip computer-aided design," *IEEE MTT-S Int. Microwave Symp. Dig.*, 1980. [verify pages]

F. Tools and data

22. ◆ C. Wolf, "Yosys Open SYnthesis Suite." [Online]. Available: <https://yosyshq.net/yosys/> — Logic synthesis for all area figures in Section VIII.
23. ◇ A. Arsenovic *et al.*, "scikit-rf: An open source Python package for microwave network creation, analysis and calibration," *IEEE Microwave Magazine*, 2022. [verify volume, issue, pages]
24. ◆ SkyWater Technology and Google, "SkyWater Open Source PDK (sky130)." [Online]. Available: <https://github.com/google/skywater-pdk> — BSIM4 device models, mismatch corners, and the sky130_fd_sc_hd standard-cell library whose LEF and SPICE views supply every area and transistor count in Table III.
25. ◆ "Ngspice, the open source SPICE circuit simulator." [Online]. Available: <https://ngspice.sourceforge.io/>
26. ◇ L. W. Nagel and D. O. Pederson, "SPICE (Simulation Program with Integrated Circuit Emphasis)," Univ. California, Berkeley, Memo ERL-M382, 1973. [verify memo number]

G. Statistics

27. ◇ J. A. Hanley and A. Lippman-Hand, "If nothing goes wrong, is everything all right? Interpreting zero numerators," *JAMA*, vol. 249, no. 13, 1983. [verify pages] — The rule of three, used wherever a run produces zero errors (Section V-B).

R. Citations still required

The following claims in this draft are *not yet* supported by a verified citation and must not be submitted in this state.

- A. **Section II-D, the evaluation-practice characterization.** Three to six read, recent neural-equalizer papers. This is load-bearing: the paper's gap statement (Section II-F) claims an omission in existing practice, and an unsupported claim of omission is the weakest possible foundation for a contribution.
- B. **Section II-E, the pre-FEC operating point.** The specific clause of 802.3ck (or 802.3bs) that fixes the target, so that 2×10^{-4} is attributed rather than asserted. Our derived 2.262×10^{-4} should be compared against the standard's own stated figure, and any discrepancy explained rather than averaged away.
- C. **Section II-C, Volterra and DA lineage** — entries [10], [11], [14], [18].
- D. **Section VIII-B, DSP-slice capacity.** The 7-series device data sheet for the DSP48E1 count used in the parallelism argument of Section XII, which is presently stated as an unmeasured hypothesis and must either be measured or cited.
- E. **Prior art on lookup-based taps.** Reported implementations of FFE taps as ROM lookups in high-speed receivers. If such work exists in the form we believe, Proposition 1's novelty is in the *compliance-referred evaluation* and the exactness argument, not in the use of a table, and the introduction must say so.

Note on novelty

Proposition 1 is not deep; the inverse of a memoryless invertible function being memoryless is immediate, and lookup tables in receivers are old. The contribution claimed here is not the table. It is the *evaluation*: that giving the baseline this free, exact inverse, judging at a threshold derived from the deployed code rather than at a chosen BER, and pricing in standard cells, jointly reverse the conclusion that a comparison omitting all three reaches. Section R-E is therefore the citation that most directly threatens the framing, and it should be resolved first.